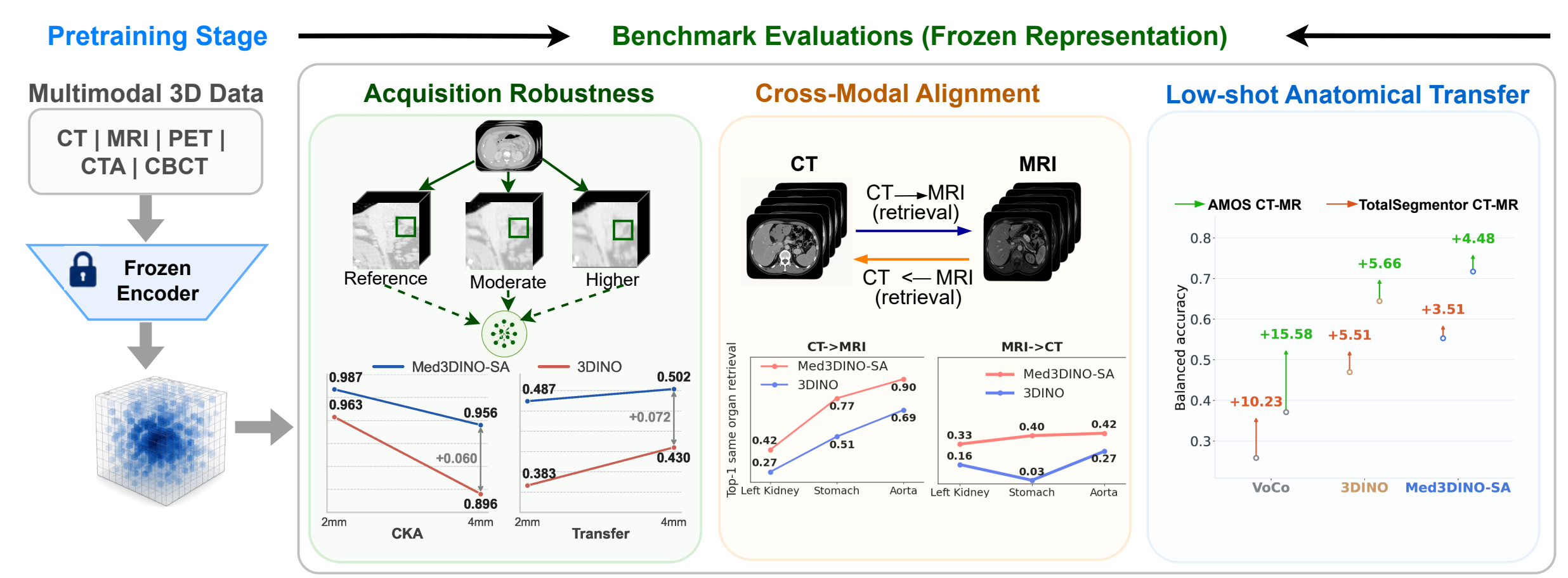

Pretraining Stage
Benchmark Evaluations (Frozen Representation)
Multimodal 3D Data
CT | MRI | PET | CTA | CBCT
Frozen Encoder
Acquisition Robustness
Reference
Moderate
Higher
Med3DINO-SA
3DINO
0.987
0.963
0.956
+0.060
0.896
2mm
4mm
CKA
0.487
0.502
+0.072
0.383
0.430
2mm
4mm
Transfer
Cross-Modal Alignment
CT
MRI
CT—→MRI (retrieval)
CT ←— MRI (retrieval)
CT->MRI
MRI->CT
Top-1 same organ retrieval
Med3DINO-SA
3DINO
0.42
0.27
0.77
0.51
0.90
0.69
Left Kidney
Stomach
Aorta
0.33
0.16
0.40
0.03
0.42
0.27
Left Kidney
Stomach
Aorta
Low-shot Anatomical Transfer
AMOS CT-MR
TotalSegmentor CT-MR
Balanced accuracy
0.8
0.7
0.6
0.5
0.4
0.3
+10.23
+15.58
+5.51
+5.66
+3.51
+4.48
VoCo
3DINO
Med3DINO-SA